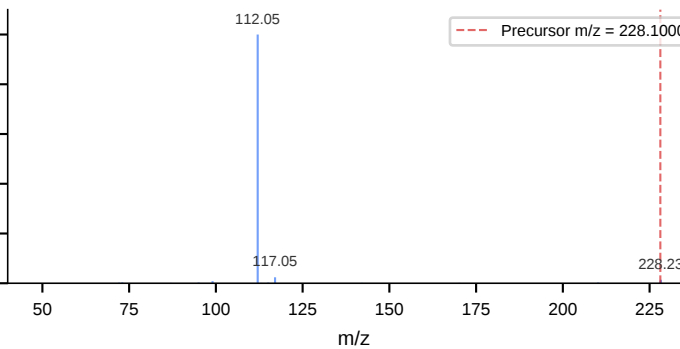


a Spectrum 1 of 10 (sub6a-gnps-CCMSLIB00000577924...) positive mode, [M+H]¹⁺, 37 peaks



All 10 spectra in this task:

#	spectrum_id	precursor m	n peaks
1	sub6a-gnps-CCMSLIB00000577924	228.1000	37
2	sub6a-gnps-MoNA031994	229.0819	41
3	sub6a-gnps-CCMSLIB00000577907	115.0500	60
4	sub6a-gnps-MoNA036841	129.0659	33
5	sub6a-gnps-CCMSLIB000005720659	226.0830	48
6	sub6a-gnps-MoNA033382	129.0659	36
7	sub6a-gnps-MoNA032862	118.0611	34
8	sub6a-gnps-MoNA036203	118.0611	30
9	sub6a-gnps-CCMSLIB000006121641	129.0660	30
10	sub6a-gnps-CCMSLIB00010142384	228.0980	67

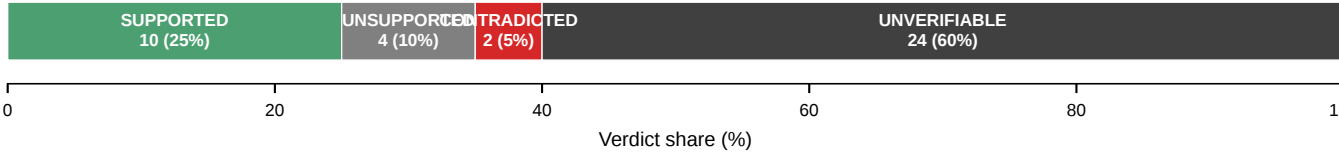
b Stage 1 — Library identification (library_search, 10/10 = 100% top-1 in 1.2s)

#	Hit	predicted (InChIKey)	ground truth (InChIKey)	predicted compound name	score
1	OK	CKTSBUTUHBMZGZ	CKTSBUTUHBMZGZ	2'-deoxycytidine CollisionEnergy:102..	0.9998
2	OK	MXHRCPNRJAMMIM	MXHRCPNRJAMMIM	HMDB:HMDB00012-21 Deoxyuridine	0.5108
3	OK	OIVLITBTBDPEFK	OIVLITBTBDPEFK	DIHYDROURACIL	0.6005
4	OK	NBAKTGXDIBVZOO	NBAKTGXDIBVZOO	5,6-DIHYDRO THYMINE	0.9062
5	OK	CKTSBUTUHBMZGZ	CKTSBUTUHBMZGZ	ReSpect:PT201440 2'-Deoxycytidine[dC..	0.6796
6	OK	NBAKTGXDIBVZOO	NBAKTGXDIBVZOO	5,6-DIHYDRO THYMINE	0.9705
7	OK	BPMFZUMJYQTVII	BPMFZUMJYQTVII	GUANIDINEACETIC ACID	0.9875
8	OK	BPMFZUMJYQTVII	BPMFZUMJYQTVII	GUANIDINEACETIC ACID	0.9295
9	OK	NBAKTGXDIBVZOO	NBAKTGXDIBVZOO	Dihydrothymine - 40.0 eV	0.9988
10	OK	CKTSBUTUHBMZGZ	CKTSBUTUHBMZGZ	2'-deoxycytidine	1.0000

c1 Identified compounds (deduplicated)c2 LLM narrative (claude-opus-4-7, 17.3s, 2738 chars)

1. 2'-deoxycytidine CollisionEnergy:102.. 2. HMDB:HMDB00012-21 Deoxyuridine 3. DIHYDROURACIL 4. 5,6-DIHYDRO THYMINE 5. GUANIDINEACETIC ACID	## Pathway Analysis ### 1. Affected Pathways The metabolite profile strongly implicates pyrimidine degradation as the primary affected pathway, with a secondary signal in creatine/arginine metabolism. Four of the five metabolites are pyrimidine-related: - 2'-Deoxycytidine — a pyrimidine deoxyribonucleoside - Deoxyuridine — a pyrimidine deoxyribonucleoside - Dihydrouracil — a pyrimidine degradation intermediate - 5,6-Dihydrothymine — a pyrimidine degradation intermediate Guanidinoacetic acid is the sole outlier, belonging to the creatine biosynthesis pathway (arginine + glycine → guanidinoacetate, catalyzed by AGAT). ### 2. Key Drivers The key drivers are dihydrouracil and 5,6-dihydrothymine. These are direct products of dihydropyrimidine dehydrogenase (DPD/DPYD), the rate-limiting enzyme in pyrimidine catabolism. Their co-elevation alongside the upstream nucleosides (deoxyuridine, deoxycytidine) suggests either increased flux through the degradation pathway or impaired downstream clearance (e.g., reduced dihydropyrimidinase activity). Deoxyuridine accumulat ... [truncated]
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d Stage 3 — Verifier verdict over 40 extracted claims



Ground truth pathway
Pyrimidine catabolism
(reactome: R-HSA-73621)

Signal compounds:
C00526, C00906, C00881, C05100, C00429

Stage 1 → Stage 3 takeaway

- 10/10 spectra correctly identified (100% top-1)
- Identified compounds match GT signals (deoxycytidine, deoxyuridine, dihydrouracil, ...)
- LLM said 'pyrimidine degradation' — CONTRADICTED by Layer 6a (non-canonical name vs RaMP top-3)
- Bottom-line: biology right, naming off

e Sample claims (8 of 40 extracted)

#6	SUPPORTED	Dihydrouracil is a pyrimidine degradation intermediate Evidence: RaMP analytehaspathway confirms compound (RAMP_C_000000053, derived from cpd:C00429) is in at least one of the 5 resolved pathways for the claim (4 membership row(s)).
#8	SUPPORTED	Guanidinoacetic acid belongs to the creatine biosynthesis pathway Evidence: RaMP analytehaspathway confirms compound (RAMP_C_000219492, derived from cpd:C00581) is in at least one of the 5 resolved pathways for the claim (3 membership row(s)).
#1	CONTRADICTED	The metabolite profile strongly implicates pyrimidine degradation as the primary affected pathway Evidence: Claimed pathway 'metabolite profile strongly implicates pyrimidine degradation' is not in top_pathways[10]; ground-truth top-1 is 'Pyrimidine catabolism'.
#2	CONTRADICTED	The metabolite profile implicates creatine/arginine metabolism as a secondary affected pathway Evidence: Claimed pathway 'arginine metabolism' is not in top_pathways[10]; ground-truth top-1 is 'Pyrimidine catabolism'.
#7	UNSUPPORTED	5,6-Dihydrothymine is a pyrimidine degradation intermediate Evidence: Pathway 'Dihydrothymine is a pyrimidine degradator' not present in this task's ground_truth_pathway, top_pathways[10], or RaMP. Treated as unsupported (no positive evid...
#9	UNSUPPORTED	The creatine biosynthesis pathway involves the reaction arginine + glycine → guanidinoacetate Evidence: Pathway 'The creatine biosynthesis pathway' not present in this task's ground_truth_pathway, top_pathways[10], or RaMP. Treated as unsupported (no positive evidence) rat...
#3	UNVERIFIABLE	Four of the five metabolites are pyrimidine-related Evidence: Layer 6a found no pathway ID or name in the claim, and no top_pathways canonical name appears verbatim in the claim text. Cannot map to a verdict.
#4	UNVERIFIABLE	2'-Deoxycytidine is a pyrimidine deoxyribonucleoside Evidence: Layer biological_sub6 found no pathway phrase / ID and no concrete entity to verify. Free-text biological role claims are out of v0 scope.